Energy-Aware Task Allocation for Cooperative Lunar Exploration

Prototype Design and Evaluation Protocol

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***Abstract*: This draft describes a two-rover prototype for cooperative exploration of a lunar test field. Rovers detect visual scientific markers, share locations, allocate inspections, recover unfinished work after a rover becomes unavailable, and return to their bases. A nearest-rover policy and an energy-aware heuristic share the same battery feasibility constraints. The implementation includes typed ROS 2 interfaces, a generated Gazebo world, camera-based marker perception, mission telemetry, and a portable rehearsal of the coordination logic. Comparative Gazebo experiments remain pending. This document presents the design and protocol without claiming measured policy improvements.**

***Index Terms*: multi-robot systems, lunar exploration, task allocation, ROS 2, Gazebo, energy**

I. Introduction

Cooperative exploration requires both distributed observations and decisions about which robot should execute each task. Multi-robot task allocation provides a framework for separating missions, tasks and robot capabilities [1]. Here, two rovers search separate regions and exchange detections. An inspection can be assigned to a rover other than the observer when travel cost and energy availability favour that decision.

The research question is whether including remaining energy in allocation improves target completion and safe return compared with a distance-only policy. The contribution is an integration and an experimental comparison of explicit heuristics, not a claim of a new optimal allocator or flight-ready lunar autonomy.

II. Related Software and Scope

The Space ROS lunar pole exploration demo provides a relevant reference with a VIPER-inspired rover, Mons Mouton terrain, cameras, solar-generation plugins and a rechargeable battery [2], [3]. These sources are recorded at fixed revisions. The baseline uses lightweight models. An experimental adapter prepares the upstream rover geometry and steerable joints with namespaced Gazebo controllers. Physical validation, detailed terrain and solar integration remain pending.

The target stack is ROS 2 Jazzy and Gazebo Harmonic, connected through `ros_gz`. OpenCV performs colour segmentation. Ground-truth odometry removes localisation uncertainty from the first comparison. The current design cannot support conclusions about SLAM, arbitrary rough terrain, real mineral identification, or physical solar energy management.

III. Mission and Architecture

The mission comprises deployment, sector exploration, visual discovery, allocated inspection and return to base. Uniquely coloured spheres represent five points of scientific interest. The rovers have different routes and initial battery fractions. Success requires every target to be confirmed and both rovers parked at their respective bases.

A. ROS Components

Each rover has a custom agent and camera detector. A central coordinator receives shared detections and status, publishes leased inspection goals and records telemetry. The agent produces velocity commands, tracks its battery, confirms nearby targets from local camera observations and handles returning and charging. A monitoring node publishes RViz markers and paths. A local mission console shows camera feeds, routes and events, with pause, resume and return controls.

Typed messages separate observations, resource state, ownership and completion. Sensor and command topics are namespaced; mission topics are shared. All mission coordinates use the world frame.

B. Visual Perception

HSV masks and contour filters detect five marker colours. Small, clipped or strongly non-circular blobs are rejected. For marker diameter D, focal length f_x, observed width w, horizontal centre u, and principal point c_x, the planar estimate is

$$z_c \approx \frac{f_x D}{w}, \qquad y_c \approx -\frac{(u - c_x) z_c}{f_x}. \tag{1}$$

The camera offset and closest timestamped rover pose transform the estimate into world coordinates. This approximation assumes a mostly level field. Detection error must be measured separately against simulator truth.

An inspection requires the assigned rover to reach its standoff, face the marker and receive recent local observations over a dwell period. A teammate's report alone cannot confirm a task.

C. Control

Agents follow sector waypoints and sweep their cameras at survey stops. The 80 by 60 m scene includes mapped rocks, outpost structures and crater exclusions. Bounded planar A* inflates these hazards, shared marker estimates and peer poses by the rover footprint and a safety margin. A forward laser scan also detects unlabelled obstacles. A command gate checks curved motion and braking distance; missing range data stops motion. Opposing rovers commit to passing on the right. Inspections use fixed stand-off approaches and recent local camera observations. These safeguards target the supplied level scene; rough-terrain traversability and general dynamic-obstacle navigation still require further work.

IV. Allocation and Energy

Each target has at most one owner, and each rover receives at most one inspection at a time. Unique target IDs suppress duplicates. Leases distinguish current assignments from expired work, and every reassignment receives a new task ID.

Let d_{ij} be rover i's approximate distance to target j's stand-off, h_{ij} its estimated return distance, b_i its battery fraction and r a reserve. The synthetic energy estimate is

$$e_{ij} = k_d(d_{ij} + h_{ij}) + k_t \left(\frac{d_{ij} + h_{ij}}{v_{\mathrm{ref}}} + t_s \right). \tag{2}$$

Both policies require $b_i \geq e_{ij} + r$. The baseline minimises d_{ij}. The energy-aware policy minimises

$$J_{ij} = d_{ij} + \lambda \frac{e_{ij} + r}{\max(b_i, \epsilon)}. \tag{3}$$

The recorded weight λ has distance units. This greedy heuristic does not solve global scheduling and may increase distance.

Energy decreases with measured distance and simulation time. Recharging at a base is accelerated for practical demonstrations. These fractions are not calibrated to a real rover. A* detours may exceed the straight-line estimate, so the agent also checks its own return budget and may release work.

A. Recovery

A missing heartbeat, unavailable or returning rover, or task timeout releases ownership. Failed tasks enter a cooldown and have bounded attempts. Results for an expired task ID are ignored. A service disables and restores a rover for repeatable failure interventions.

V. Evaluation Protocol

Run matched experiments with the same world, initial state, camera settings, mission duration, resource parameters and failure schedule. Only the policy changes. Scenarios should include nominal exploration, asymmetric batteries, temporary rover failure, and the combined energy/failure case.

Retain failed and timed-out runs. Report completion time alongside success rate instead of averaging only successful runs. Use distinct predefined layouts; record and pair random seeds if noise or randomisation is introduced. Retain software revisions, configuration, CSV telemetry, JSONL events and hardware details with each run.

TABLE I
Metrics required from actual Gazebo runs

Metric	Definition
Confirmed targets	Unique locally confirmed markers by deadline
Mission success	All targets confirmed and both rovers returned
Completion time	Simulation time to mission success
Distance	Sum of measured rover paths
Energy	Synthetic consumption; state units explicitly
Recovery latency	Failure to reassignment and completion
Detection error	Estimated location versus simulator truth

A. Validation Boundary

Portable tests check ownership, feasibility, leases, recovery, detours and synthetic marker images. An offline rehearsal imports the same mission logic, but uses kinematic motion and oracle detections. It does not validate wheel contact, rendering, camera-based world positions or physical lunar operation. A Linux gate separately checks rendered cameras, laser scans, odometry, detections, assignments and conservative footprint separation during a compact mission. Geometric clearance is not a measurement of physical contact forces.

VI. Results: Pending Gazebo Experiments

The policy comparison remains pending. Report hardware and software versions, paired-run counts, success rates, the metrics in Table I, and a recovery timeline from actual Gazebo runs. Offline rehearsal values are debugging evidence only. Equal performance or an energy/time trade-off is a valid finding.

VII. Discussion and Limitations

Ground-truth localisation, known-size coloured markers, planar control and an accelerated battery limit external validity. Report sensitivity to the heuristic weight and error in the return-budget estimate. The central coordinator is a single point of failure. Further work includes physical validation of the detailed rover adapter, uneven terrain, solar charging, terrain-aware navigation and richer perception.

VIII. Conclusion

A prototype and comparison protocol have been prepared for cooperative lunar exploration. Conclusions about the allocation policies remain pending actual Gazebo comparisons.

References

[1] B. P. Gerkey and M. J. Matarić, *A formal analysis and taxonomy of task allocation in multi-robot systems*, International Journal of Robotics Research, vol. 23, no. 9, pp. 939-954, 2004, doi: 10.1177/0278364904045564.

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